

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-19 04:45 UTC

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Generated 2026-05-19 04:45 UTC

- Cycles recorded: **824**
- Time span: 607.1h (~1.36 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	784
BARRIER_HIT	20
FALSIFIED	16
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	40
Schur-Weyl Duality	28
Free Probability Theory	19
Representation Theory of Symmetric Groups	18
Matroid Theory	17
Free Probability	14
Additive Combinatorics	13
Noncommutative L^p Geometry	10
Random Matrix Theory	8
Polymatroid Theory	8
Algebraic Geometry	7
Spectral Graph Theory	7
Finite Geometry	6
Plethysm Theory	6
Persistent Homology	5
FOURIER_ANALYSIS	5
Noncommutative Harmonic Analysis	5
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4

Field	Cycles
Fourier Analysis on Boolean Functions	4
Algebraic Geometry over Finite Fields	4
Fourier Analysis of Boolean Functions	4
Algebraic Geometry of Secant Varieties	4
Noncommutative Geometry	4
Fourier analysis of boolean functions	3
Algebraic Topology	3
matroid theory	3
Coarse Geometric Lifting (CGL) — Geometric Complexity Theory	3
{“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”}	3
Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)	3

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-19 02:13 UTC	INCONCLUSIVE	Lorentzian Defect of Cut Polynomial Lower-Bounds
2026-05-19 02:22 UTC	INCONCLUSIVE	Max-CUT SoS-2 Ga Level-1 Fourier Mass Fraction of Term Family Bounds
2026-05-19 02:31 UTC	INCONCLUSIVE	Monotone CLIQ Dilworth Width of Gate-Cone Poset Lower-Bounds AC^0 PARITY Size
2026-05-19 02:40 UTC	INCONCLUSIVE	Mean Log-Bias of Internal Gates Lower-Bounds AC^0 PARITY Size
2026-05-19 02:48 UTC	INCONCLUSIVE	Sandpile Group Exponent Lower-Bounds Tseitin
2026-05-19 02:53 UTC	INCONCLUSIVE	Tree-Resolution Size SAW Connective Growth of Gate DAG Lower-Bounds
2026-05-19 03:06 UTC	INCONCLUSIVE	$ACC^{0[2]}$ for MOD_3 Frobenius Fourier Defect of Barrington AND_n PBP Trajectory
2026-05-19 03:17 UTC	INCONCLUSIVE	Eigenvalue Gap in SOS Moment Matrices for Max-CUT Approximation
2026-05-19 03:33 UTC	INCONCLUSIVE	Matroid Rank Complexity of Monotone DNF for k-CLIQUE
2026-05-19 03:40 UTC	INCONCLUSIVE	Communication Complexity Lower Bound for AC^0 PARITY via Real Alge
2026-05-19 03:56 UTC	INCONCLUSIVE	Log-Size Submodularity in Monotone DNF for k-CLIQUE
2026-05-19 04:03 UTC	INCONCLUSIVE	SOS Moment Matrix Eigenvalue Threshold for ACC^0 Lower Bounds

Time	Verdict	Title
2026-05-19 04:17 UTC	INCONCLUSIVE	Fourier Coefficient Concentration in ACC ⁰ Circuits
2026-05-19 04:24 UTC	INCONCLUSIVE	Schur-Weyl Multiplicity Gap in Symmetric vs Antisymmetric Decompo
2026-05-19 04:45 UTC	INCONCLUSIVE	Communication Complexity Lower Bound for AC ⁰ Circuits via Real Po

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (sudo apt install pandoc texlive-xetex), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```